



A novel approach to Predict WTI crude spot oil price: LSTM-based feature extraction with Xgboost Regressor

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ABSTRACT

This paper presents a novel model based on LSTM to predict future prices of WTI crude oil. The WTI price forecasting utilizes data on spot gold price, US 10-year bond yield, global economic activity, and US dollar index from January 1986 to May 2023. The model's performance is assessed using measures such as MAE, MSE, RMSE, MAPE, and R^2 metrics. The results generated by the proposed new model are compared to those of the existing machine and deep learning methods, and it is observed that the new model performs better than the existing models in all statistical tests. The study further examined the decision-making processes of the model using SHAP analysis and assessed the individual contribution of each feature to the model's predictions. The correlation between the US Dollar Index and Gold prices and WTI crude oil prices is evident. The SHAP research has demonstrated that the model effectively captures complicated economic linkages and enhances the accuracy of forecasts. The results of this study enhance the development of models that are capable of predicting results, even in times of significant instability, such as economic crises. Using sophisticated data analytics and AI methods would improve the efficiency of energy market oversight.

1. Introduction

In addition to being an energy source, crude oil is also an economic commodity. Particularly when it is assumed that the demand for commodities will increase during periods of dollar decline, crude oil becomes an important actor in economies beyond its conceptual dimension [1]. The price of crude oil is determined by supply and demand according to market conditions. However, in some cases, price volatility occurs when effective conditions are taken into account [2,3]. Investors, policymakers, and energy professionals benefit from accurate crude oil price forecasts. In this situation accurate prediction of crude oil price is important. Recently, it has become difficult to predict the price of crude oil due to the turmoil in the global energy market.

The correlation between oil prices and economic and financial indicators has consistently been a subject of intense debate. Achieving a complete understanding of this relationship and generating precise predictions would represent a crucial advancement in determining the policies to be implemented in the future. It is well recognized that volatility in oil prices has various effects on economic indicators. For example, fluctuations in oil prices might impact inflation, leading to shifts in the supply curve, or affect the balance of payments, thereby

influencing exchange rates. Given the significance of the correlation between crude oil prices and currency rates, several researchs have been conducted in the literature to explore this relationship [4]. The importance of the United States of America (USA) dollar as a significant reserve currency and the use of dollars for oil trading increases the interconnection between these two assets [5,6]. The exchange rate is an important factor for predicting the price of crude oil. Gold is often believed to have a significant correlation with crude oil, making it a crucial instrument. Gold has the largest trading volume among metals, whereas oil has the biggest trade volume among basic materials. Within this paradigm, the assumption that crude oil and gold exhibit a correlation in the commodities market justifies the need of monitoring the price of gold. There is a notion that as oil prices rise, the amount of gold in a country's reserves also increases, resulting in a correlation between oil and gold. This implies that the present location and value of gold are influential in determining the price of oil [7,8].

Bond yields are a key predictor of crude oil prices in addition to the correlation between gold and oil. The United States possesses the most extensive bond market globally. As previously stated, the status of the US as the reserve currency results in a significant and enduring impact on the bond market. When examining the correlation between the US

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ten-year oil price and bond rates, it is unavoidable to assume that they exhibit a synchronized movement [9]. Finally, the real economic activity index is integrated into the model to improve the effectiveness of the determining factors. The relationship between global economic activity and oil prices is well-established, as fluctuations in economic activity directly impact the demand for crude oil [10–12]. Accurately forecasting the oil price, together with its related factors, will provide guidance for the design of policy. Standard statistical models and time-series analysis cannot fully capture the complex dynamics and non-linear patterns of crude oil price fluctuations.

Considering these constraints, Deep Reinforcement Learning (hereafter, DRL) may be a promising method for crude oil price forecasting. The development of DRL in artificial intelligence allows agents to find optimal strategies by interacting with the environment and receiving rewards or penalties for their actions. DRL methods have been shown to outperform conventional models and adapt to market dynamics in financial forecasting. DRL research on crude oil price prediction is the focus of this study. We aim to provide valuable insights into the efficacy, drawbacks, and potential benefits of using DRL algorithms to accurately predict crude oil prices by examining recent developments in this field.

For this purpose, firstly, when the existing literature is examined, multitude of studies that have examined the use of artificial intelligence (AI), specifically deep learning methods, in the context of forecasting crude oil prices [2,13–32]. Among the various AI methodologies, neural networks have gained significant popularity particularly Long Short-Term Memory (hereafter, LSTM) networks. This popularity stems from their remarkable capacity to effectively capture non-linear patterns and temporal dependencies within time series data.

Building on this foundation, numerous research studies have also utilized Artificial Neural Network (ANNs) of various configurations, such as Multilayer Feed Forward and Recurrent Fuzzy Neural Networks, to achieve accurate predictions of crude oil prices [13,17–19,22–24,30]. These models have outperformed more conventional econometric models in terms of performance, successfully identifying the intricate non-linear correlations found in crude oil data.

Prior researches has investigated the integration of AI methodologies, specifically ensemble empirical mode decomposition (EEMD) [26, 29,33] and LSTM [31–37] networks in order to increase the accuracy of forecasting. The integration of EEMD and LSTM demonstrated encouraging outcomes in addressing the non-stationary and nonlinear attributes of crude oil prices.

Furthermore, many researchers used various models and integrated additional data, including economic indicators, news sentiments, and social media data, into the AI models in order to enhance the precision of predictions [2,14,16,20,21,25,27,28,38,39]; [41]; [42–45]. The integration of external factors has demonstrated promise in capturing the influence of exogenous events and sentiment on fluctuations in crude oil prices.

Liang et al. [46] seek to create an innovative DRL algorithm for predicting crude oil prices in three prominent commodity exchanges. The algorithm will focus on forecasting prices multiple steps ahead. They use Brent, Western Texas Intermediate (WTI), and Oman data sets. But crude oil price has other determinants [47]. explored that convenience yield (CY), 10 year bond yields, and global economic activity are the main determinants of value of crude oil [48]. stated that the main determinants of crude oil value are crude oil price (spot), Organization of Petroleum Exporting Countries (OPEC) reference basket price, futures oil price, dollar index, gold price and economic policy uncertainty index [46]. stated that to ensure the price forecast accuracy, researchers should consider external factors to predict the crude oil price. In this study, we forecast the crude oil price using the above mentioned main determinants of crude oil price such as WTI crude oil spot price, US Dollar index, global real economic activity, gold price and 10 years BY to make the forecast more accurately. At the same time, monthly data between January 1986 and May 2023 were used by expanding the data set in our study.

The main difference of this study from the previous literature is that it estimates the price of crude oil using LSTM and eXtreme Gradient Boosting (XGBoost) algorithms. The major contribution of the study is that the applied model is used for the first time in the estimation of crude oil, and in this context, the existing literature will be expanded. The purpose of developing this model is to enhance the understanding of the temporal characteristics and complicated relationships within time series data, ultimately leading to improved accuracy in predicting. LSTM models are particularly differentiated by their ability to capture extended dependencies and complicated trends in time series models. The LSTM model's feature extraction enables more effective learning of long-term relationships and complicated structures within the dataset. Furthermore, the LSTM model's identified notable characteristics enhanced the predictions of a robust deep learning method like XGBoost. This enhances the capacity of the proposed model to make accurate predictions and generalize effectively. The conclusion part of the research presents a comparison between the prediction performance of our proposed model compared to other models discussed in the literature. It is evident that our model outperforms competing models, offering superior outcomes. In this context, an extended literature review will be presented in the continuation of the study. Then, the data set will be explained and the model methodology will be stated. The findings will be evaluated in the next section. Finally, the conclusion and policy recommendations will be given.

2. Literature review

Determining the price of crude oil has always been an interesting topic. Especially in recent times, many studies have been developed to determine the price of crude oil accurately through different models or analyses. The literature has extensively examined ANNs as potent instruments for forecasting crude oil prices. Numerous investigations have illustrated ANNs' efficacy in apprehending intricate market dynamics. In this case, the studies carried out to determine the crude oil price by applying ANNs are as follows;

In a study by Suriya [13], the application of ANNs was employed to forecast Brent crude oil values. The results revealed that the implementation of a model featuring 200 neurons in the hidden layer yielded a mean absolute percentage error (MAPE) of 1.98 %. This outcome demonstrates the model's notable accuracy in predicting crude oil prices. In a similar way, Kulkarni and Haidar [18] put forward an ANN model to forecast the direction of immediate future cost of crude oil. Their model demonstrated a range of directional accuracy, varying from 53 % to 78 %, when predicting price movements one to three days in advance. The results of their study suggest that ANN models have the capability to offer significant insights for traders and investors operating in the short-term crude oil market. Pan, Haidar, and Kulkarni [17] utilized artificial neural networks as a means to capture and represent the underlying implicit nonlinearity inherent in the forecast of crude oil prices. The Brock-Dechert-Scheinkman test provided evidence of the existence of non linear dynamic phenomena in crude oil prices. Additionally, the proposed ANN model demonstrated an out-of-sample accuracy rate of approximately 80 %, indicating its potential as a valuable predictive tool for future price trends. Sompui and Wongsinlatam [22] added another model to the analysis. They conducted a comparative analysis of various ANN models and the least square method (LSM) in order to assess their effectiveness in forecasting the trend of short-term crude oil spot prices. The research findings emphasized the superiority of ANN models, specifically the model with four hidden layers, in comparison to LSM in accurately forecasting fluctuations in crude oil prices. Abdullah [23], by improving standart ANN models, examined the application of machine learning models in the forecasting future prices of crude oil. The models investigated included a hierarchical conceptual (HC) model, an artificial neural network-quantitative (ANN-Q) model, a Linguistic model, a Rule-based Expert model, and a Hybridisation of linguistic quantitative (LQ) model. The research underscored the

efficacy of back-propagation trained neural network (BPNN) within the ANN-Q framework, highlighting its ability to produce the results that are both more accurate and competitive. Hamdi and Aloui [24] conducted a comprehensive review of the previously published studies related to the prediction of crude oil prices through the use of ANNs. The authors emphasized the success of ANNs in effectively capturing complicated inverse correlations between variables used as inputs and those used as outputs. The review offered valuable insights into the various approaches and methodologies utilized in forecasting crude oil values through the use of ANN based models. In a more recent context, Gupta and Nigam [30] conducted an investigation into the ability to predict of ANNs with different configurations of hidden layers in determining the direction of where the price of crude oil is moving. The findings of their research demonstrated that the ANN model, specifically one with four hidden layers, achieved the highest level of accuracy. This highlights the significance of carefully selecting the architecture of the model in order to effectively forecast price direction. Furthermore, Xu et al. [45], utilized regression to evaluate the efficacy of econometric and ML models in predicting the daily WTI crude oil price. Traditional econometric models like ARIMA, GARCH, and OLS were evaluated against ANN regression and ANN time series models to determine the best method and model (econometric or machine learning). Their study suggested modifying ANN hidden layer bias and weights to improve prediction performance. They highlighted, neural networks outperformed econometric models in comparison.

While many researchers use ANN, different neural models, such as empirical mode decomposition (EMD), EEMD, adaptive particle swarm optimization (APSO), multiple wavelet recurrent neural network (MWRNN), recurrent fuzzy neural network (RFNN), Elman recurrent-neural network (ERNN) etc., are preferred, too. In their study, Yu, Wang, and Lai [14] introduced a novel approach that combines EMD using ensemble learning from neural networks to forecast future prices of crude oil. The empirical findings illustrated the accuracy of their methodology, which integrated neural networks that can adapt and learn to combine the forecasting results of intrinsic mode functions and achieve an ensemble output. This approach offers an efficient means of forecasting values of crude oil. Li et al. [26] improved the standart EMD model. They presented a novel approach that combines EEMD, APSO, and relevance vector machine (RVM) to forecast crude oil values. The research conducted by the authors emphasized the favorable effectiveness of the EEMD-APSO-RVM approach in projecting future prices of crude oil, demonstrating superior performance in terms of root mean squared error (RMSE), MAPE, and Dstat measurements.

In their study, Haidar, Kulkarni, and Pan [15] implemented a comprehensive investigation into the prediction of prices for crude oil in the near future. They employed a three-layer feedforward neural network for this purpose. The study underscored the network's capacity to effectively process noisy time series data with a high degree of precision, provided that it is appropriately designed and trained. This finding underscores the network's potential for real-world implementation within the energy market. The study conducted by Alexandridis and Livanis [16] examined the utilization of wavelet neural networks in the prediction of WTI crude oil spot prices. The researchers compared the performance of their model with the traditional econometric forecasting approach. The study conducted by the researchers showcased the efficacy of the suggested wavelet neural network model in capturing the multiscale data characteristics of crude oil prices. Furthermore, it was observed that the model surpassed the performance of the traditional econometric model. Similarly, Mingming and Jinliang [49] proposed the utilization of a MWRNN model as a means of analyzing crude oil prices. The model demonstrated an impressive degree of prediction accuracy by integrating wavelet analysis and RNN forecasting. The testing data exhibited an average error rate of 4.06 %, while the training data displayed a slightly lower average error rate of 3.88 %. These findings offer valuable insights that can be used by traders and risk managers. Wang and Wang [50] proposed an innovative neural

network framework that integrates multilayer perceptron (MLP) and ERNN to predict fluctuations in crude oil prices. The proposed architecture demonstrated exceptional accuracy and efficiency in forecasting fluctuations in crude oil prices, thereby offering a valuable resource for traders and investors operating within the energy market. The study conducted by Abdullah and Zeng [19] demonstrated the effectiveness of SVMs in predicting crude oil values. The researchers utilized essential economic indicators as input features and the value of crude oil as the output variable. The study conducted by the researchers emphasized the significant precision of SVMs in predicting crude oil prices, thus positioning them as a viable and promising substitute for conventional statistical and econometric methodologies. In their empirical investigation, Hu, Hu, and Lin [20] researched the application of a RFNN for the goal of making predictions about the prices of crude oil futures. The findings of their study demonstrated that the RFNN model exhibited superior performance compared to other neural networks used in the analysis. This suggests that the RFNN model possesses enhanced predictive capabilities in accurately capturing the complex structure of crude oil futures prices. Shin et al. [21] investigated the efficacy of SSL in forecasting volatility of crude oil prices. To identify the existence of nonlinearity in the price dynamics, the researchers employed the Brock-Dechert-Scheinkman test. The study placed particular emphasis on the benefits of SSL in depicting the network connections among time-series entities, resulting in enhanced precision when evaluating the influence of economic factors on crude oil prices. In 2016, Mahdiani and Khamehchi presented a neural network model that was optimized using genetic algorithms. To better forecast the price of crude oil, this model set out to do just that. Proving its resilience across diverse market conditions, the model managed with limited input data or significant variable fluctuations. Chen et al. [27] predicted crude oil prices using deep neural networks and deep belief networks. The authors found that deep learning models can capture the complex and non-linear aspects of crude oil price fluctuations, improving price prediction accuracy. Zhao and Yu [2] put forth a methodological framework that uses a combination of SDAE and bootstrap aggregation to predict crude oil prices using deep learning techniques. The experimental findings indicated that the ensemble model exhibited superior performance compared to individual models, thereby highlighting possible improvements to the accuracy of crude oil price forecasts using deep learning ensembles. Gao and Lei [28] forecast crude oil prices using stream learning. This method captures oil price dynamics while reducing computational costs. The authors found that stream learning can predict volatile crude oil prices. The study showed stream learning can incorporate real-time data updates. Jahanshahi et al. [44] developed a forecasting the price of crude oil application using machine learning and deep learning. The researchers estimated prices using historical price data and external factors, providing energy market traders and risk managers with valuable insights. Another neural network model is used to improve forecasting accuracy with artificial intelligence. Amal [43] presented findings that highlight the superior predictive capabilities of the LSTM model when combined with the Adaptive Moment Estimation (Adam) optimization technique. Specifically, the study focused on forecasting Crude Palm Oil (CPO) prices. [41] proposed a new method based on online media text mining to predict crude oil prices. Using convolutional neural network (CNN), hidden patterns are extracted from news texts and these patterns are shown to be related to price fluctuations. The main findings of the study reveal that the proposed topic-sensitivity synthesis prediction models outperform the old benchmark models. The metrics used include MAE and RMSE. According to the MAE and RMSE values, the SVR model performed the best with all selected features. The combination of text features and financial features provided significantly higher accuracy in crude oil price prediction compared to financial features alone. In particular, in the random forest model, the combination of text and financial features provided a larger percentage improvement than the financial features. This study reveals that automatically extracting text features from online news using deep learning algorithms and

demonstrating the explanatory power of these features in crude oil price prediction is an important contribution to the literature. Lin et al. [51], the performance of the hybrid model based on wavelet transform with the combination of BiLSTM, Attention mechanism and CNN proposed for the prediction of crude oil futures prices is evaluated with various metrics. The models used include BiLSTM, BLAC (BiLSTM-Attention-CNN), CNN-LSTM, CNN, CNN, CNN-BiLSTM, GRU, LSTM, BiLSTM-CNN, LSTM-CNN, BP (Back Propagation) and SVR (Support Vector Regression). RMSE, MAE, MAPE and R^2 were used as performance metrics. The proposed WT-BLAC model outperformed all other models in RMSE, MAE, MAPE and R^2 metrics. For example, the WT-BLAC model outperformed all other models in WTI price forecasting with RMSE: 2.2518, MAE: 1.1780, MAPE: 2.6261 and R^2 : 0.9663 in the WTI price forecast, while in the Brent price forecast, RMSE: 1.2936, MAE: 0.8894, MAPE: 1.8027 and R^2 : 0.9900, providing the highest accuracy. Compared to other models, the WT-BLAC model exhibits superior performance in terms of prediction accuracy and error rate, and is able to make more accurate predictions by reducing noise using wavelet transform. These results show that the proposed model is more effective than other methods in forecasting crude oil futures prices and has the potential to improve forecasting accuracy, especially in financial markets. The LSTM model demonstrated superior performance in comparison to conventional statistical models, thereby highlighting its potential in effectively predicting commodity prices. Zhang and Hong [31] found similar results when they compared LSTM models' crude oil forecasting accuracy across temporal scales. When compared to other models, the LSTM model reliably predicted higher prices for crude oil, according to their study. This shows the model's flexibility and ability to model crude oil market dynamics. Li et al. [32] conducted an effective forecasting approach incorporating trajectory similarity is proposed by using the TS and LSTM. The methodology can effectively represent the attributes of crude oil price, which are not linear. According to their study, VMD-SE-TS/LSTM excelled all benchmark models, including hybrid models with decomposition. Crude oil price forecasting works well with VMD-SE-TS/LSTM. In summary, the collective body of research suggests that ANN, deep learning techniques, and other machine learning approaches provide useful resources for predicting crude oil prices, exhibiting varied levels of precision. The accurate prediction of outcomes is greatly affected by the careful selection of model architecture, optimization strategies, and input features.

Although the relevant literature contains many studies, the accurate prediction of oil prices is still not fully achieved. In light of this, a new model has been developed in this study, unlike the current applications. It is thought that this model will produce better results than the results found in the current literature. The methodology of the relevant model will be presented in the continuation of the study.

3. Methodology

This paper introduces a novel deep learning model that utilizes LSTM and XGBoost algorithms to forecast crude oil prices. This section provides a comprehensive description of the data collection, software and hardware utilized for analysis, the estimating methods employed, and the assessment measures utilized.

3.1. Data set

Our data set includes 449 monthly data between 1986 January and 2023 May. 4 independent variables were used to predict oil prices. These are US Dollar Index, Global Economic Activity, US 10 Year Bond Yield and Gold Price data. The sources from which these data were obtained are shown in Table 1.

Basic descriptive statistics of the dataset are presented in Table 2 below.

Table 1

Summary and source of the variables.

Variable	Date Range	Source
WTI Crude Oil	Jan 1986–May 2023	https://www.eia.gov
US Dollar Index	Jan 1986–May 2023	investing.com
Index of global real economic activity	Jan 1986–May 2023	dallasfed.org
10 Year Bond Yield for USA	Jan 1986–May 2023	st.louisfed.org
Gold Price	Jan 1986–May 2023	investing.com

Table 2

Descriptive statistics.

Variable	Mean	Median	Std Deviation	Min	Max
US Dollar Index	92,53428	92,73	10,1889692	71,8	121,52
Index of global real economic activity	0,903341	−6,69	58,43485309	−161,62	189,2
10 Year Bond Yield for USA	4,6898	4,5	2,297008187	0,62	9,52
Gold Price	800,1799	450,48	536,0638474	255,95	1991,34
WTI Crude Oil	46,38751	36,75	29,51279102	11,35	133,88

3.2. Data Pre-processing

The dataset utilized for estimating crude oil prices was suitably prepared and standardized prior to analysis. This section provides a comprehensive explanation of the data preprocessing steps. Preliminarily, the data set was examined for any instances of missing data and anomalies. There is a complete absence of missing data. To enhance the efficiency and accuracy of data processing, all variables were standardized using the Min-Max Scaling approach. This strategy normalizes the value of each variable within the range of 0 and 1, hence enhancing the model's learning speed and stability. The Min-Max Scaling method has been implemented according to formula 1 provided:

$$x_{\text{normalized}} = \frac{x - x_{\min}}{x_{\max} - x_{\min}} \quad (1)$$

The variable x represents the original data value, whereas $x_{\min}$ and $x_{\max}$ reflect the minimum and maximum values of the relevant variable. Following the normalization process, the data set was partitioned into separate training and test sets in order to assess the models' performance. For this study, 80 % of the data set was allocated for training purposes, while the remaining 20 % was reserved for testing. Thus, the training dataset consisted of 359 data points, while the testing dataset contained 90 data points. The training set was utilized to train the models, while the test set was employed to assess the models' overall performance. The use of these preprocessing processes ensured the precise processing of the data, leading to enhanced effectiveness of the prediction models.

The study employed the Python 3.11 programming language for data manipulation and model creation. The data manipulation and analysis tasks were performed using the Pandas and NumPy libraries. Data visualization was done using the Matplotlib and Seaborn libraries. Machine learning models and evaluation metrics were implemented using the Scikit-learn library. Deep learning models were developed using the TensorFlow and Keras libraries. Lastly, reinforcement learning algorithms were implemented using the Stable Baselines library. Analyses were conducted on a system equipped with an Intel Core i7-9700K CPU (3.6 GHz, 8 cores), NVIDIA GeForce RTX 2080 graphics processing unit (8 GB GDDR6), 32 GB DDR4 memory, and 1 TB NVMe SSD storage in order to fulfill high-performance computing needs. The Windows 11 Pro operating system was utilized. The utilization of this software and

hardware infrastructure facilitated the efficient and successful execution of the training and testing procedures for crude oil price prediction models.

3.3. Forecasting models

This study utilized six separate advanced machine learning and deep learning algorithms to forecast crude oil prices: DDPG (Deep Deterministic Policy Gradient), LSTM (Long Short-Term Memory), SVR (Support Vector Regression), CNN (Convolutional Neural Network), XGBoost, and a novel deep learning model based on LSTM-XGBoost. Each of these strategies will now be detailed in a comprehensive manner.

3.3.1. DDPG

The mentioned algorithm is a hybrid of Deep Q-learning and Policy Gradient. As seen in Fig. 1, the architecture consists of two DNN, two target networks, an environment (E), and a buffer store. Model parameters and hyperparameters are all part of the algorithm. The weights of the DNN (ϕ^Q , ϕ^μ , $\phi^{Q'}$ and $\phi^{\mu'}$) serve as approximators for functions, while the variables characterize the learning procedure, including the training of the DNNs. The role of hyperparameters in determining how an algorithm learns is substantial. Consequently, it is imperative to modify these parameters to achieve optimal learning performance. The subsequent section provides a description of the hyperparameters utilized in conjunction with DDPG's educational procedure. Fig. 1 indicates the steps of DDPG algorithm. The subsequent sections provide an explanation of each component inside the DDPG framework.

3.3.1.1. Q network and target network. The critic DNN is built on top of Q-learning, a reinforcement learning algorithm that Watkins and Dayan [52] introduced as a value-based framework. The critic, known as DNN or Q-Network, consists of a Q value function represented as $Q(s_t, a_t, \phi^Q)$ and model parameters represented by ϕ^Q . The critic does a mathematical calculation to determine the ideal Q value function, denoted as ϕ^μ , in order to identify the most favorable action, referred to

as $\alpha^\mu(s_t)$.

$$\alpha^\mu(s_t) = \arg\max_a \phi^\mu(s_t, a_t) \quad (2)$$

The Q value can be characterized as the expected reward following the successful completion of a single action, serving as a metric for evaluating the efficacy of that action in relation to the identification of the optimal strategy μ . Identifying the most efficient actions for distinct spaces is a relatively uncomplicated task. This study considers action spaces as continuous, which poses a greater challenge in optimizing the problem. The potentially infinite action space makes it impossible to calculate the maximum operator over the actions in Equation (3). In order to address this concern, it is hypothesized that $Q^\mu(s_t, a_t)$ possesses differentiability in relation to the actions a_t . This property allows for the utilization of a policy $\mu(s_t)$ based on gradients in Equation (3), denoted as Equation (4).

$$\max_a Q^\mu(s_t, a_t) \approx Q(s_t, \mu) \quad (3)$$

The gradient-based policy $\mu(s_t)$ will be further examined in the actor-based DNN procedure. The Bellman equation describes the optimal-value function $Q^\mu(s_t, a_t)$.

$$Q^\mu(s_t, a_t) = E[r(s_t, a_t) + \gamma \max_{a_{t+1}} Q^\mu(s_{t+1}, a_{t+1})] \quad (4)$$

The participant initiates the process through the estimation of a function that is close to the optimal value $Q^\mu(s_t, a_t)$ using the given recursive equation. The critic network, also known as the Q network, receives updates through the computation of the loss function, which is denoted as the MSBE:

$$L = \frac{1}{T} \sum_t [y_t - Q(s_t, a_t | \phi^Q)]^2 \quad (5)$$

where:

$$y_t = r(s_t, a_t) + \gamma Q'(s_{t+1}, \mu'(s_{t+1} | \phi^{\mu'}) | \phi^{Q'}) \quad (6)$$

The variable τ represents a randomly selected subset of departures from the buffer for replays. The main purpose of the replay buffer is to

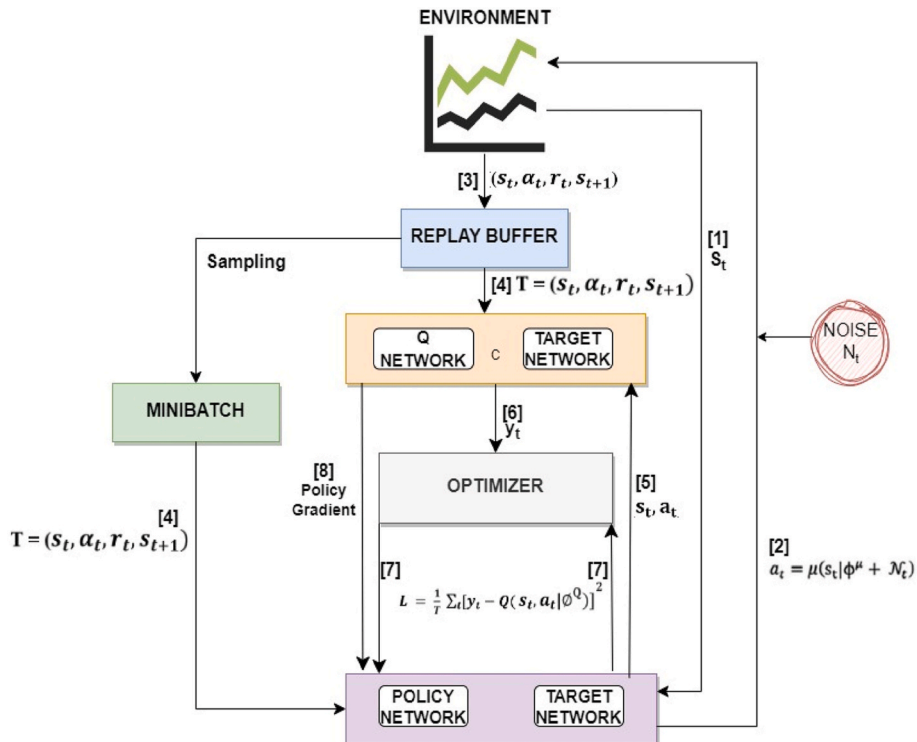


Fig. 1. Ddpq algorithm structure.

preserve a log of previous encounters, as explained in the following section. The target value is denoted by y_t , while both the target critic network and the actor-critic network produce Q and μ' as their output. The incorporation of target networks is a crucial advancement inside the DDPG algorithm. Target DNNs refer to replicated versions of the actor and critic networks that are intentionally delayed. This delay is implemented to enhance the stability of the algorithm and prevent any potential divergence. As given in Fig. 1, The Q-network and Policy network's model parameters are passed on to the Target DNNs. (step 8 in Fig. 1). However, in order to prevent the target DNNs from relying on the same model parameters ϕ , A "soft update" method is used to periodically refresh the target DNNs (Equation (7)). This entails gradually copying over the parameters ϕ^Q, ϕ^μ of the DNNs that are intended to be used by the Q-network and Policy network.

$$\phi^Q + (1 - \tau) \phi^Q \rightarrow \phi^Q \quad (7)$$

$$\phi^\mu + (1 - \tau) \phi^\mu \rightarrow \phi^\mu$$

The hyperparameter τ is responsible for determining the frequency at which the target networks' weights are changed. Typically, τ falls within the range of 0.001–0.01. The significance of incorporating a soft update mechanism lies in its ability to prevent instability in the algorithm by addressing the issue of a moving target. Devoid of any lag time for the desired DNNs in equation (5), the MSBE would continuously pursue a target that is always changing, resulting in an unstable algorithm. As seen in Equation (6), the inclusion of target DNNs μ' and Q' is essential for the computation of the target action value y_t . Hence, Minimizing the loss function L is the goal of the algorithm in equation (5) in order to reduce the disparity between the Q-value function $Q(s_t, a_t | \phi^Q)$ and the target value y_t , which takes into account the Q value for the subsequent state (s_{t+1}). This process, known as TD learning, involves minimizing the error between the current time step's Q-value function and The anticipated Q-value function derived from the target DNNs. The DDPG algorithm optimizes this error by updating the critic network using stochastic gradient descent. An important part of the update process is the learning rate, which controls how much the model changes based on the TD error. A low learning rate may cause separation as a consequence of the limited adaptability of the model, whereas a high learning rate can lead to unstable training or inappropriate parameter values.

3.3.1.2. Buffer memory. A replay buffer is a storage mechanism utilized by an agent to efficiently retain its experiences. Its primary objective is to maintain a record of a finite number (P) of past experiences, represented as tuples $\{s_t, a_t, r_t, s_{t+1}\}$. As the quantity of collected experiences reaches the maximum capacity of P tuples, the oldest experiences are subsequently removed from the buffer. To help the algorithm develop a stable behavior, during training it is sampled from the accumulated experiences using random minibatches of size T . The optimal amount of the buffer is dependent upon the nature of the problem at hand, as a shortage of data may result in overfitting, while a surplus of data may impede the learning process. In order to reduce algorithmic divergence, it is important that the data utilized during the training phase exhibits independent distribution. Hence, in each iteration, the DNNs receive updates through taking a random sample of a minibatch from a buffer. In order to lessen the impact of any existing relationships between the data, this method is used.

3.3.1.3. Policy network. At each time step t , the policy network, also called the actor network, observes an environmental state and continues to take action. Similarly to other RL algorithms, the main objective of the agent is to identify the most optimum strategy. The DDPG algorithm utilizes a policy gradient approach to identify the most optimal policy. This is achieved by employing a parameterized function, denoted as $\mu(s | \phi^\mu)$, which deterministically translates a given state to a corresponding action. The substantial computational burden of computing

the state distribution's gradient can be avoided by employing a deterministic strategy. The policy's performance gradient is equal to the deterministic policy gradient, according to previous research [53]. The derivative of the policy may be estimated by utilizing the chain rule, as seen in Equation (8). This allows for the subsequent update of the actor-network and actor-target network parameters, which are ϕ^μ and $\phi^{\mu'}$, through the utilization of gradient ascent. The primary purpose in this context is to maximize the reward.

$$\begin{aligned} \nabla_{\phi^\mu} J &\approx E_{s_t} [\nabla_{\phi^\mu} Q(s, a | \phi^Q) | s = s_t, a = \mu(s | \phi^\mu)] \\ &= E_{s_t} [\nabla_a Q(s, a | \phi^Q) |_{s=s_t, a=\mu(s | \phi^\mu)} \nabla_{\phi^\mu} \mu(s | \phi^\mu) |_{s=s_t}] \end{aligned} \quad (8)$$

In order to facilitate the process of exploration, a Gaussian-type process known as OU noise [54] is incorporated by introducing variable N_t before returning the action to the environment (Eq. (9)). The observed noise might be viewed as a discrepancy between the plant and model, leading to the sampling of random noise from a normal distribution with correlations.

$$a_t = \mu(s_t | \phi^\mu + N_t) \quad (9)$$

Liang et al. [55] have reported that zero-mean Gaussian noises are considered appropriate for DDPG algorithm. From the perspective of process control, this particular characteristic has appeal due to the inherent complexity and nonlinearity of the majority of issues, which often necessitate computationally intensive methods when using traditional software. The DDPG algorithm stands out due to its model-free nature, which means that the probabilities of transitions are unknown. In other words, the agent lacks prior knowledge about the functioning of the environment. This property becomes advantageous in situations when no a priori information is available. As illustrated in Fig. 1, the methodical stages of the DDPG learning process per episode can be briefly described as follows [56].

- The recorded data presents the states of the model as they were initially.
- The policy network is responsible for receiving the state as input and generating an action as output. Before returning the action to the environment (as shown in Equation (9)), a sort of Gaussian process known as OU noise is included. The noise appears as an indication of the discrepancy between the plant and model in the context of accounting.
- Based on the activity performed, the environment is provided with a reward r_t and generates the next state s_{t+1} . The elements of the MDP, denoted as (s_t, a_t, r_t, s_{t+1}) , are kept in the buffer memory.
- A minibatch of T transitions $((s_t, a_t, r_t, s_{t+1}))$ is randomly sampled from the buffer memory.
- The critic network is provided with the state s_t and action a_t .
- The critic network determines the target y_t as shown in Equation (6).
- The critic is updated by minimizing the loss function L , which is calculated based on the TD error as described in Equation (5).
- The actor-network can be updated by taking the derivative of the policy, as shown in Equation (8).
- equation (7) represents the updating process for target networks.

The learning process concludes after a certain number of episodes are completed by the user, or when a specific ending criterion set by the user is met, such as the absence of improvement in rewards across a significant number of episodes.

3.3.2. Long short-term memory (LSTM)

LSTM is able to study and predict major events in time series with long time intervals because of its unique design [57]. A gating mechanism controls the flow of information in LSTM, a recursive neural network. This gating mechanism uses a basic recursive neural network [58–60]. The LSTM model is composed of the input, output, and forget

gates [61]. LSTM algorithm steps are in Fig. 2. The procedure for acquiring an LSTM unit is as outlined below:

Within the distinct cellular architecture of the LSTM model, the forget gate assumes the role of selectively eliminating the cell state from the preceding sequence. The prior time series hidden state, denoted as h_{t-1} , and the current time series input, denoted as x_t , are processed by the activation function σ_g to produce the output vector f_t of the forget gate. This relationship can be mathematically represented as Equation (10):

$$f_t = \sigma_g (W_f x_t + U_f h_{t-1} + b_f) \quad (10)$$

The weight coefficients associated with the forget gate are denoted as W_f and U_f , while the bias coefficient is represented by b_f . The activation function is indicated by σ_g .

The input gate controls the transmission of input data into the currently active cell. The coefficients i_t and C'_t in this gate are computed based on the hidden state h_{t-1} from the previous time step and the current time series input x_t . These coefficients determine the proportion of information to be updated and preserve the cell candidate information. The computation of these coefficients necessitates the utilization of the activation function.

$$i_t = \sigma_g (W_i x_t + U_i h_{t-1} + b_i) \quad (11)$$

$$C'_t = \sigma_c (W_c x_t + U_c h_{t-1} + b_c) \quad (12)$$

The weight coefficients are denoted as W_i , U_i , W_c , and U_c . The bias coefficients are represented by b_i and b_c , while the activation functions are indicated by σ_g and σ_c (Equation (11) and Equation (12)).

The cell state receives an update by a linear interaction process, wherein the prior cell state is modified and subsequently included into the current cell state, so completing the information update. Furthermore, the cell state, denoted as C_t , is updated by multiplying the output of the forget gate, f_t , with the previous cell state, C_{t-1} , and adding the product of the output of the input gate, i_t , and the cell candidate data, C'_t . This calculation results in the new cell state, C_t (Equation (13)).

$$C_t = f_t \times C_{t-1} + i_t \times C'_t \quad (13)$$

The output gate is responsible for the computation and transmission of information to the subsequent time step. The output vector o_t is derived by applying the activation function σ_g to the input vectors h_{t-1} and x_t (Equation (14)).

$$o_t = \sigma_g (W_o x_t + U_o h_{t-1} + b_o) \quad (14)$$

The cell state weight coefficients, denoted as W_o and U_o , and the bias coefficient, denoted as b_o , are associated with the input gate. Subsequently, the output gate produces the output o_t , which is then multiplied

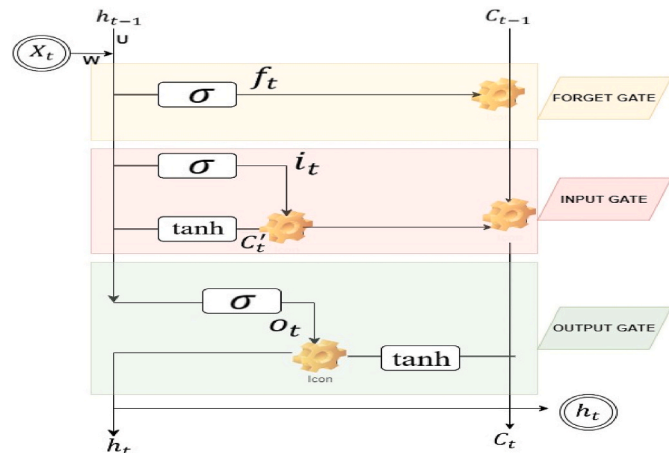


Fig. 2. The structure of LSTM.

by the current sequence cell state C_t . This product is then processed by the activation function $\tan h$ to generate the final output of the hidden layer (Equation (15)).

$$h_t = o_t \times \tan h (C_t) \quad (15)$$

Explanation of the principal phases of the LSTM forecast model follows [62].

- To begin with, the external state at the previous moment (h_{t-1}) and the input vector at the current moment (x_t) are utilized in the computation of three gates and the candidate state C'_t .
- Secondly, the forget gate (f_t) and the input gate (i_t) are combined to update the internal state (C_t) of the present moment.
- At the last phase, using the output gate (o_t), the internal state information is conveyed to the outside state h_t .

3.3.3. Support vector machine regression (SVR)

SVM methodology was initially proposed by Ref. [63]. Subsequently, a regression technique known as SVR [64] was developed.

Gao et al. [65] state that the purpose of SVR is to ascertain a linear regression function, denoted as $f(x)$, within a high-dimensional space.

$$f(x) = \omega \Phi(x) + b \quad (16)$$

Suppose that x is a term of the set of real numbers, denoting the sample vector. The function Φ is a non-linear mapping.

The implementation of a loss function for linear insensitivity, denoted as $L(f(x), y, \epsilon)$, assists to improve the reliability of the optimization problem. This loss function can be mathematically represented as follows:

$$L(f(x), y, \epsilon) = f(x) = \begin{cases} 0 & |y - f(x)| \leq \epsilon \\ |y - f(x)| - \epsilon & |y - f(x)| > \epsilon \end{cases} \quad (17)$$

To determine the forecast of ω and b , it is necessary to solve the following optimization problem:

$$\begin{cases} \text{Min. } \frac{1}{2} \|\omega\|^2 + C \sum_{i=1}^n \xi_i \\ \text{sub. t. } \begin{cases} y_i - \omega \Phi(x_i) - b \leq \epsilon + \xi_i \\ -y_i + \omega \Phi(x_i) + b \leq \epsilon + \xi_i^* \\ \xi_i, \xi_i^* \geq 0 \end{cases} \end{cases} \quad (18)$$

Where $x_i \in \mathbb{R}^d$ and $y_i \in \mathbb{R}$ be the output value and input vector containing the serial number i , accordingly. Here, d represents the number of elements in an input vector. The variable n represents the quantity of training samples. The symbol ϵ represents the measure of regression precision. The variable C is a penalty factor that quantifies the degree of penalty imposed on a data sample when its error exceeds the threshold value ϵ . The variables ξ_i and ξ_i^* serve as slack variables in order to impose penalties on the complexity of the fitting parameters.

The optimization problem can be resolved using a Lagrange function, which gives a formula for the resulting solution:

$$\omega^* = \sum_{i=1}^l (\alpha_i - \alpha_i^*) \Phi(x_i) \quad (19)$$

$$b^* = \frac{1}{N_{sv}} \left\{ \sum_{0 < \alpha_i < C} \left[y_i - \sum_{x_j \in SV} (\alpha_i - \alpha_i^*) K(x_i, x_j) - \epsilon \right] + \sum_{0 < \alpha_i < C} y_i - \sum_{x_j \in SV} (\alpha_j - \alpha_j^*) K(x_i, x_j) + \epsilon \right\} \quad (20)$$

The variable N_{sv} represents the quantity of support vectors that have been determined. The Lagrange multipliers, denoted as α_i and α_i^* , are

non-negative. The kernel function, denoted as $K(x_i, x_j)$, is utilized in this context. Specifically, the Gaussian kernel function, which exhibits superior generalization capability, is chosen.

$$K(x_i, x_j) = \exp\left(-\frac{\|x - x_i\|^2}{2\sigma^2}\right) \quad (21)$$

The ultimate regression function is represented by Equation (21):

$$f(x) = \sum_{i=1}^l (\alpha_i - \alpha_i^*) K(x_i, x) + b^* \quad (22)$$

The process of SVR can be succinctly visualized and given in Fig. 3, which includes the input layer, hidden layer, and output layer.

3.3.4. CNN

CNN utilizes a unique linear mathematical operation known as convolution in at least one of its layers for data processing [66]. CNN networks are composed of neurons that undergo self-optimization through the process of learning. Every individual neuron carries out a specific operation using the inputs, much like how it is done in artificial neural networks. In the final layer, a loss function is derived by comparing the predicted output with the actual output. The process for CNN is similar to that of artificial neural networks. On the other hand, CNN differ from artificial neural networks in that they employ a pattern recognition area to identify the inputs in the initial stage. Specifically, when compared to standard neural networks with layers of similar size, these networks can be trained more seamlessly due to their reduced number of connections and parameters [67]. CNN is composed of various types of hidden layers [68].

- (1) In the realm of signal processing, the convolutional layer is a key player. It carries out operations on two important signals: the input signal and the filter (also known as the kernel) applied to the input. The process at hand involves the multiplication of matrices, where the input set and kernel are combined to produce a modified input. This modified input effectively extracts the necessary information from the entire input, utilizing the kernel to obtain the essential data.
- (2) The pooling layer is a method of discretization based on sampling. The goal is to decrease the dimensionality of the input feature, such as input data or feature extraction, and extract the information regarding the relationship between the input and output found within the subset.
- (3) The output layer, also referred to as the fully connected layer, establishes connections between every neuron in one layer and every neuron in the following layer.

The CNN model is applied using the following formulas:

$$x_j^d = \varnothing \left(\sum_{i \in M_j} x_i^{d-1} * w_{ij}^d + b_i^d \right) \quad (23)$$

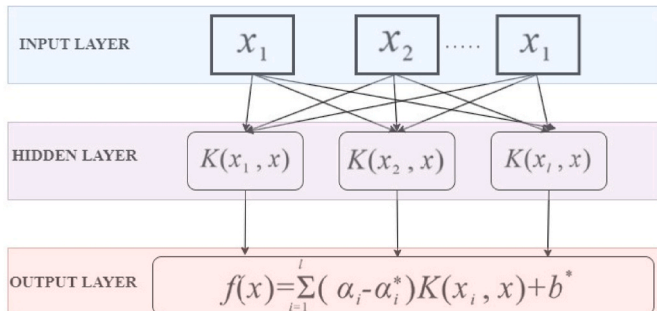


Fig. 3. Structure of SVR model.

$$x_j^d = \varnothing \left(\beta_j^d \text{down}(x_j^{d-1}) + b_i^d \right) \quad (24)$$

$$y^k = \varnothing \left(w^k x^{k-1} + b^k \right) \quad (25)$$

In Equation (23), x_j^d represents the j th feature map of the convolution layer. The feature map is vital to this topic. The activation function ($\varnothing$) is applied to the input M_j , where d is the input dimension. The layer receives features from the input feature set. A parameter w_{ij}^d represents the attribute map of the j th attribute in the d th convolutional layer. The d -1st element's attribute. Layer i references convolution layer attribute j and layer $(d-1)$ attribute i . The parameter b_i^d represents the layer-specific bias term. In Equation (24), $\text{down}(\cdot)$ is the subsampling function and β is the weight matrix. After applying the convolutional and pooling layers, Equation (25) categorizes utilizing the complete connection layer. Equation (25) explicitly defines layered neural network variables. Layer index " k " is specified. y^k is the whole link layer output. x^{k-1} is the whole link layer input. w^k represents weighting coefficient, whereas b^k represents variance.

3.3.5. XGBoost

XGBoost is a well-known method in the field of machine learning for predicting prices such as stock price forecasting [69]. Traditional Boosting Tree approaches typically rely solely on first derivative information. When incorporating residuals from the previous $n-1$ trees, distributed training for the n th tree becomes significantly more intricate. XGBoost utilizes a second-order Taylor expansion on the loss function to leverage the capabilities of CPU multithreading. XGBoost employs various techniques to mitigate the issue of overfitting, as discussed by Li, Yin, Quan, and Zhang [40]. The technique of Equations (26)–(33) provides a solution for the XGBoost algorithm. Equation (26) demonstrates the application of regression trees. K represents the total number of weak learners, while f_k denotes the number of weak learners. Equation (27) provides the goal function. Equation (27) employs a range of loss functions to address specific scenarios. Equation (27) is commonly employed to evaluate the disparity between observed and expected values $l(y_i, \hat{y}_i^{(t)})$ and the intricacy of the model $\sum_{k=1}^t \Omega(f_k)$. In order to assess the primary factor, one must replace the expected value ($\hat{y}_i^{(t)}$) with the value of the i th sample in the t th cycle. The computation utilizes the Taylor series approximation at the predicted value of y from earlier iterations ($\hat{y}_i^{(t-1)}$), as demonstrated in Equation (26). Equation (29) illustrates the primary and subsequent derivatives of the loss function $l(y_i, \hat{y}_i^{(t)})$ within the given context. Upon careful examination of the information provided, we can proceed with the computation of the derivative by substituting Equations (29)–(31) into Equation (27). Equations (32) and (33) provide the solution to the given problems. The object "Obj*" is used to quantify the numerical results of the loss function. Lower scores indicate a preference for tree structures that are idealized. The variable w_j^* provides the optimal weighting variable solution for the given situation.

$$\hat{y}_i = \varnothing(x_i) = \sum_{k=1}^K f_k(y_i), f_k \in \mathcal{F} \quad (26)$$

$$\min L^{(t)}(y_i, \hat{y}_i^{(t)}) = \min \left(\sum_{i=1}^n l(y_i, \hat{y}_i^{(t)}) + \sum_{k=1}^t \Omega(f_k) \right) \quad (27)$$

$$\Omega(f) = \gamma T + \frac{1}{2} \lambda w^2 \quad (28)$$

$$\min L^{(t)} = \min \left(\sum_{i=1}^n \left[g f_i(x_i) + \frac{1}{2} h f_i(x_i) \right] + \Omega(f_i) \right) \quad (29)$$

$$g_i = \partial_{y_i(t-1)} l(y_i, y_i^{t-1}) \quad (30)$$

$$h_i = \partial_{y_i^{t-1}}^2 l(y_i, y_i^{t-1}) \quad (31)$$

$$w_j^* = - \frac{\sum g_i}{\sum h_i + \lambda} \quad (32)$$

$$obj^* = - \frac{1}{2} \sum_{j=1}^T \frac{(\sum g_i)^2}{\sum h_i + \lambda} + \gamma \cdot T \quad (33)$$

3.4. Proposed methodology

This paper proposes a novel model that combines LSTM (Long Short-Term Memory) and XGBoost algorithms to forecast crude oil prices. The purpose of developing this model was to enhance the understanding of temporal aspects and intricate interactions within time series data, ultimately leading to improved prediction performance. In the following section, we provide a comprehensive explanation of the structure and functioning of the suggested model. Initially, the acquired data in the study were normalized using the min-max normalization method and then separated into training and test data sets. The proposed model employed the random search technique with K-Fold to optimize the model hyperparameters. By conducting tests on various segments of the data set, the model's performance and its capacity to generalize were enhanced. Subsequently, the LSTM model and k-fold cross-validation with a value of 15 were employed to perform feature extraction. The

feature extraction procedure improved the acquisition of features from the time series data, resulting in enhanced performance of the XGBoost model used in the regression step. The hyperparameters employed in the XGBoost model were calculated via the random search algorithm. Once the hyperparameters for the XGBoost stage were determined, the gathered data was utilized as the input for the XGBoost model. Fig. 4 illustrates the suggested method for determining the Crude Spot Oil Price. LSTM models are identified by their capacity to capture extended relationships and intricate patterns, particularly in time series models. The LSTM model improved the learning of long-term dependencies and complicated structures in the data set through feature extraction. Furthermore, the LSTM model's derived significant features enhanced the predictions of XGBoost, a very effective deep learning technique. It enhanced the model's capacity to generalize and improved its ability to make accurate predictions.

3.5. Evaluation metrics

Six different algorithms were applied to predict oil prices. These are SVR, LSTM, DDPG, CNN, XGBoost, and our novel LSTM-XGBoost based deep learning algorithms. For this, first of all, The collected data were divided into training and test sets in a proportion of 80:20. The data were then normalized. Afterwards, a model was created for each algorithm and trained. After the training phase was accomplished, the models were tested and the model performances were determined with the metrics RMSE, MSE, R2, RMSE and MAPE.

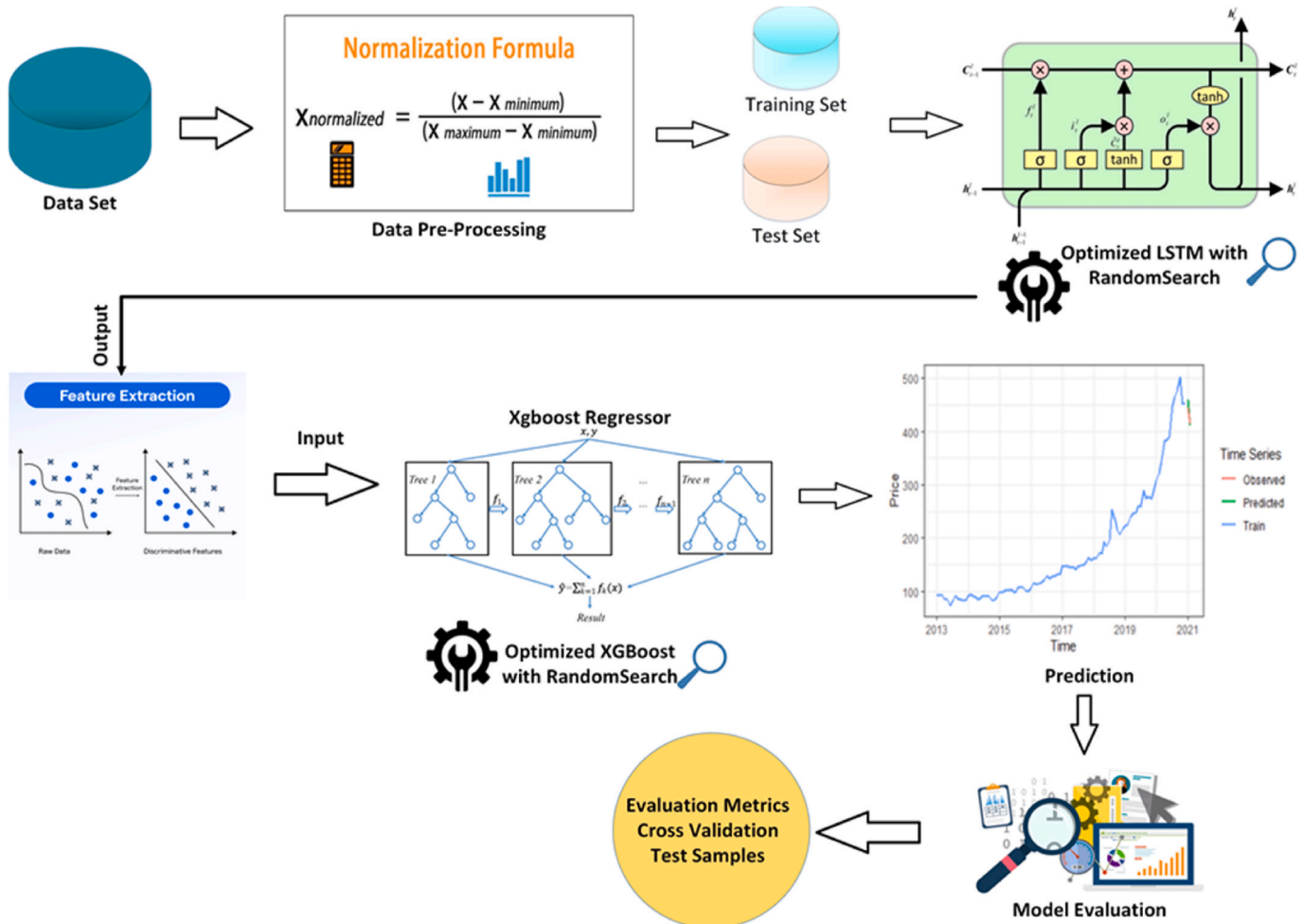


Fig. 4. Proposed model.

$$RMSE = \frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i) \quad (34)$$

$$MSE = \frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i)^2 \quad (35)$$

$$RMSE = \sqrt{\frac{\sum_{i=1}^N (y_i - \hat{y}_i)^2}{N}} \quad (36)$$

$$MAPE = \frac{1}{N} \sum_{i=1}^N \text{abs} \left(\frac{(y_i - \hat{y}_i)}{y_i + 1e - 10} \right) \quad (37)$$

$$R^2 = 1 - \frac{\sum_i (y_i - \hat{y}_i)^2}{\sum_i (y_i - \mu)^2} \quad (38)$$

4. Findings

This section presents the performance results of the proposed model and other utilized models for forecasting crude oil prices. Initially, the outcomes of the correlation matrix, which was constructed to analyze the connections among the variables in the dataset, are elucidated. Next, the results acquired by the application of time series decomposition techniques are showcased. Nevertheless, this research presents the hyperparameter settings determined by the RandomSearch algorithm for all models used. Ultimately, a thorough evaluation is conducted on the training, testing, and cross-validation outcomes of all employed models. The assessments were conducted and displayed utilizing diverse criteria to ascertain the precision and efficacy of the models. The findings offer crucial insights for selecting the most effective approach in

forecasting crude oil prices. Fig. 5 displays the correlation connections among many financial and economic data, including the US Dollar Index, real economic activity, US Bond Yield, gold, and spot crude oil.

The correlation matrix displays the relationship among different financial and economic factors, including the US Dollar Index, actual economic activity, US Bond Yield, gold, and spot crude oil. The data reveals a negative correlation (-0.42) between the US Dollar Index and real economic activity, suggesting that an increase in the dollar index is associated with a drop in economic activity. Gold and spot crude oil exhibit a robust positive correlation of 0.75 , suggesting that their prices tend to rise in tandem. The US Bond Yield and gold exhibit a strong negative connection (-0.80), indicating that when bond rates rise, gold prices tend to decline. There is a moderate negative correlation (-0.50) between spot crude oil and the US Dollar Index, suggesting that oil prices generally move in the opposite direction of the dollar's value. These findings offer crucial insights into the interconnectedness of significant economic and financial variables.

Fig. 6 presents a thorough examination of the dynamics and time series components between crude oil prices and related economic indicators. The Observed chart displays the fluctuations of the US Dollar Index, Global Real Economic Activity Index, US 10-Year Bond Yield, Gold Price, and Spot Crude Oil Price over a period of time. Gold prices have exhibited a consistent upward trend since the early 2000s and reached their highest point in the 2020s. It has been noted that some factors exhibit either more consistent or variable patterns. The Residual chart, which displays the residuals of the time series after removing the trend and seasonality, is utilized to examine the distribution of the error components of the model across time. The heightened instability, particularly during and following the 2008 financial crisis, garners significant attention. Residuals are crucial for assessing the performance of a model. The Seasonal chart is employed to comprehend the fluctuations in economic activity and prices that occur throughout specific periods of the year by illustrating the seasonal components of each variable. Gold

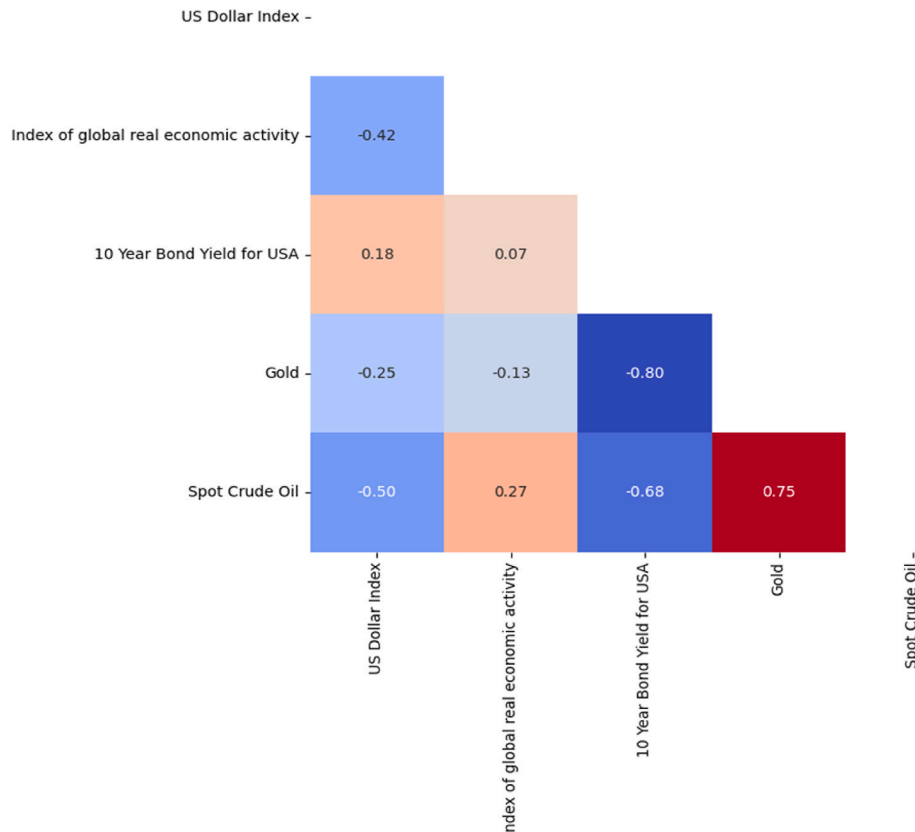


Fig. 5. Correlation matrix.

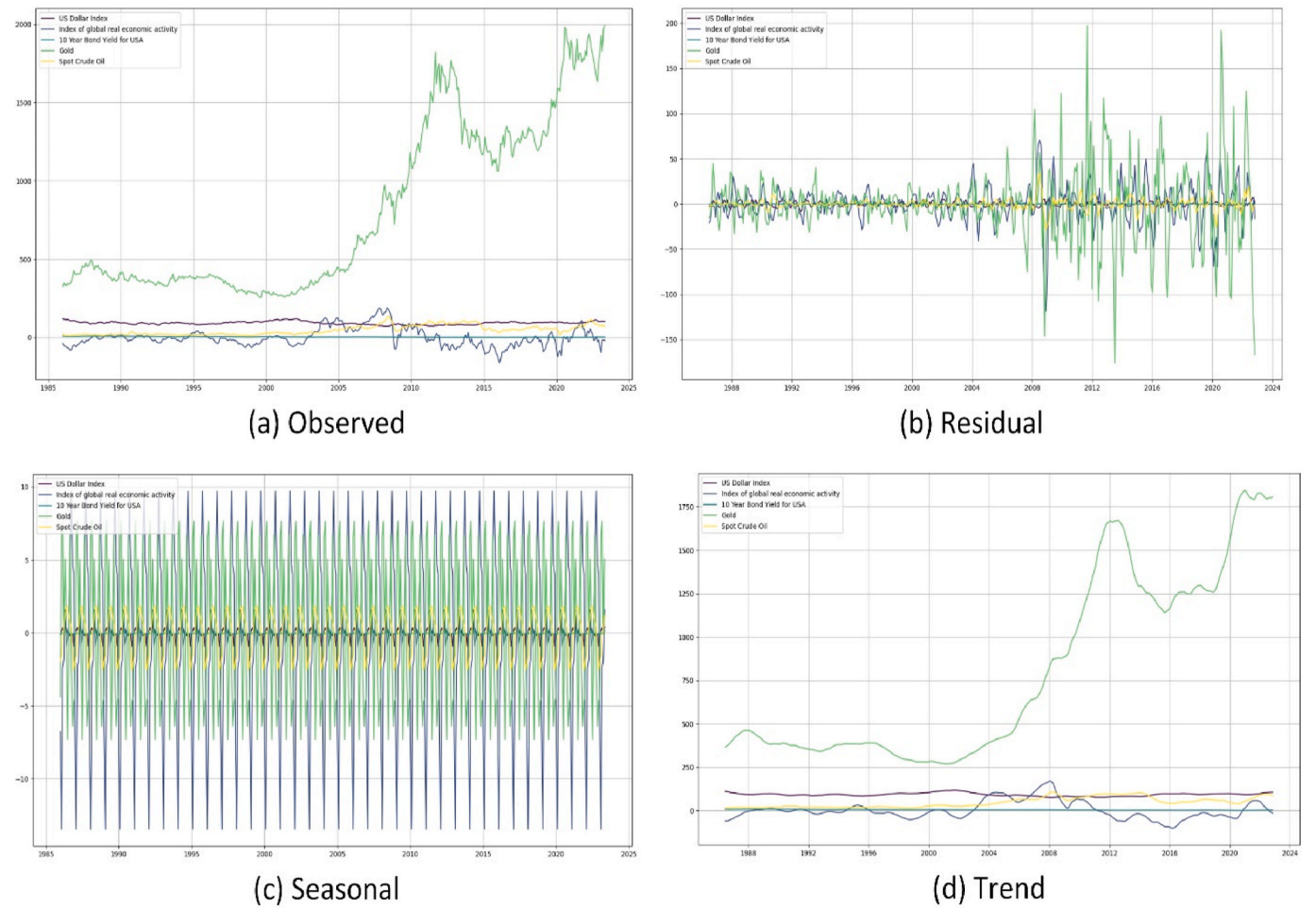


Fig. 6. Decomposition of time series components.

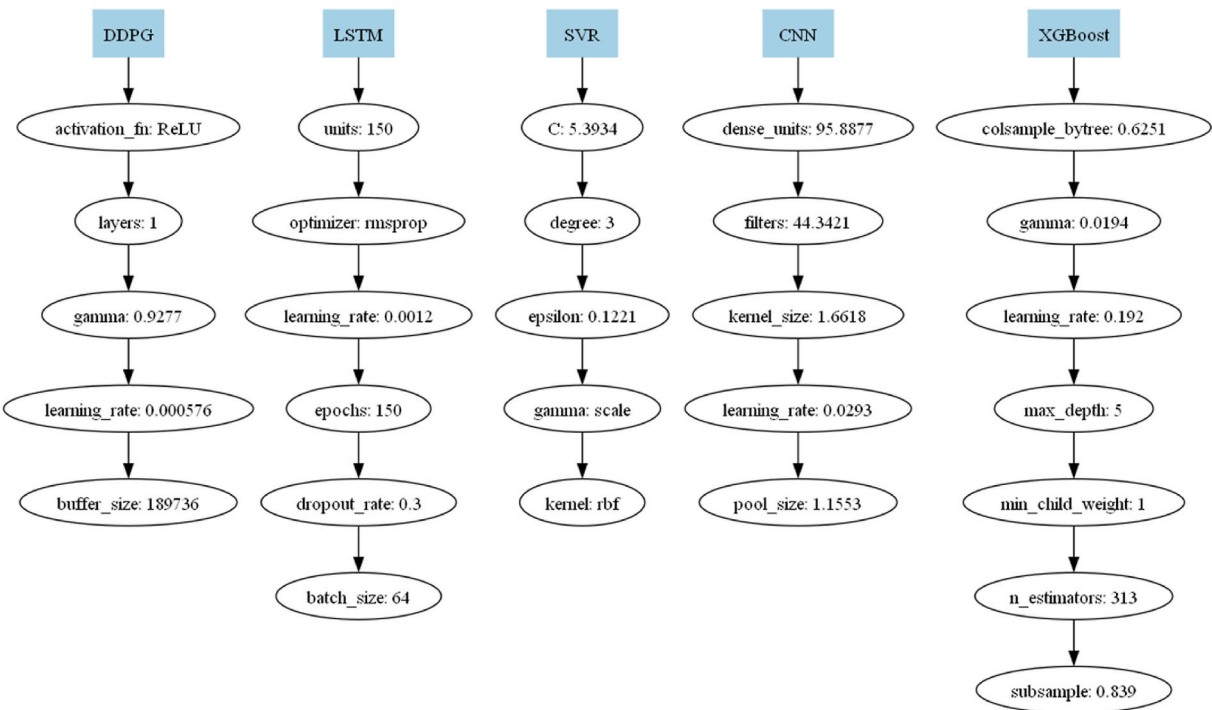


Fig. 7. Hyperparameters of the models determined by RandomSearch.

prices see a notable seasonal variation. Annual cycle patterns occur on a regular basis. The Trend chart indicates that gold prices have exhibited a consistent upward trajectory since the early 2000s, culminating in their highest point in the 2020s. It has been noted that some factors exhibit either more consistent or variable patterns.

The RandomSearch algorithm was employed to enhance the effectiveness of the models utilized in forecasting crude oil prices. RandomSearch is a highly efficient approach for finding the optimal parameter combinations by randomly sampling from a broad spectrum of hyperparameters. This approach is crucial for enhancing the precision and generalizability of models. Fig. 7 provides a comprehensive explanation of the optimal hyperparameters identified for each model and the methodology used to obtain these configurations.

Table 3 displays the R^2 , MAE, MAPE, MSE, and RMSE performance metrics for the training, testing, and 15-fold cross-validation sets of each model.

Based on the findings presented in Table 3, the DDPG model demonstrates robust performance on the training set, achieving a R^2 value of 0.9799, an MAE value of 0.0237, a MAPE value of 0.0685, an MSE value of 0.0012, and an RMSE value of 0.0345. Nevertheless, the test performance decreases to a value of 0.8660 for the coefficient of determination (R^2). Nevertheless, it demonstrates a satisfactory level of generalization with a cross-validation R^2 value of 0.9187. Similarly, while the LSTM model has strong training performance with a R^2 value of 0.9236, it experiences a notable decline in testing performance with a R^2 value of 0.8546. The cross-validation findings indicate a somewhat superior capacity to generalize compared to DDPG, as evidenced by a R^2 value of 0.9523. The SVR model has consistent performance in both the training set (R^2 : 0.9214) and the test set (R^2 : 0.8999). However, it exhibits a lower cross-validation R^2 (0.9017) compared to other models, suggesting that its capacity to generalize is limited. The CNN model exhibits a high level of performance on the training set (R^2 : 0.9177), but, there is a minor decrease in performance on the test set and cross-validation (test R^2 : 0.8498, cross-validation R^2 : 0.9312). This demonstrates the effectiveness of the CNN model on time series data, however it may have a somewhat higher error rate compared to alternative methods. The XGBoost model demonstrates excellent performance (R^2 : 0.9679) on the training set and obtains satisfactory results on the test set (R^2 : 0.8790). The cross-validation results indicate that the model has a high level of generalization ability, with a R^2 value of 0.9346. Our model demonstrates superior performance compared to other models. It achieves

great results in both training (R^2 : 0.9997) and test sets (R^2 : 0.9991), while its cross-validation R^2 indicates the highest level of generalization ability at 0.9987. This model achieves the lowest MAE, MAPE, MSE, and RMSE values across all datasets, highlighting its superior accuracy and reliability in predicting crude oil prices. The results demonstrate that the our model exhibits superior performance across all metrics and is the most appropriate model for forecasting crude oil prices. Fig. 8 displays the forecast graph for both the training and testing stages of the proposed model.

Furthermore, Fig. 9 illustrates the impact of several characteristics on the model's output of the proposed model using SHapley Additive ExPlanations (SHAP) values, both on the training and test data. The initial SHAP summary plot illustrates the influence of characteristics on the training data. The US Dollar Index exhibits a broad range of SHAP values, with higher values yielding a positive impact and lower values resulting in a negative impact. Gold demonstrates the advantageous impact of elevated values and the detrimental impact of diminished values. The Global Real Economic Activity Index exhibits a more consistent but typically lesser impact, as evidenced by a more compact distribution of SHAP values in the training data. The US 10-Year Bond Yield exhibits a positive impact when values are high and a negative impact when values are low in the training process. The SHAP values have a wide range, indicating that the model assigns considerable significance to this feature. The second SHAP summary figure illustrates the influence of several features on the model's output using the test data of the proposed model. The US Dollar Index exhibits a variable influence on model predictions, characterized by a broad range of SHAP values. Higher feature values are associated with a positive impact, whilst lower values have a negative impact. Gold demonstrates a complex yet notable impact, with high values yielding a positive influence and low levels resulting in a negative influence. The Global Real Economic Activity Index has a variable impact, as indicated by SHAP values that are closer to zero. On the other hand, the US 10-Year Bond Yield has a significant impact, with a wide range of SHAP values. Higher values indicate a positive influence, while lower values indicate a negative effect.

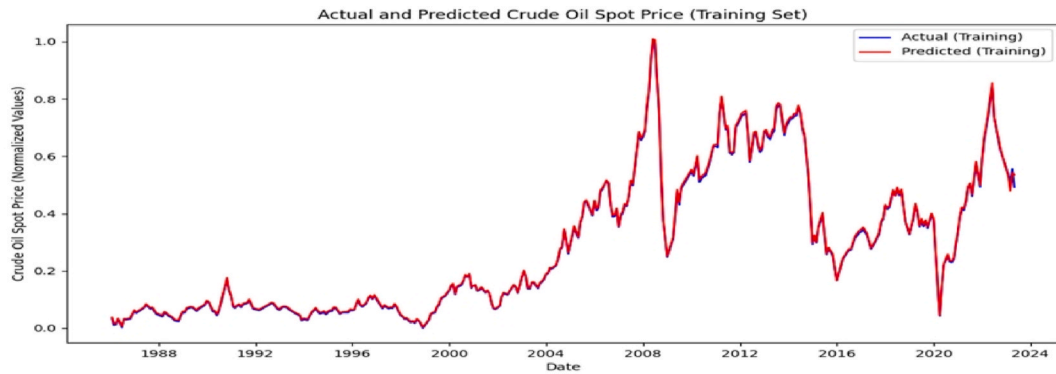
5. Conclusion, discussion and suggestions

In the study, a new model was created to predict the spot price of WTI crude oil and the analysis was performed. The obtained results show that it performs better than the existing models. We conducted an analysis of many data points, including the price of WTI crude oil, the price of gold, the yield of US 10-year bonds, global economic activity, and the US currency index. According to the obtained results, the proposed model showed superior performance in terms of the lowest MAE, MSE, RMSE and MAPE values among the training, testing and cross-validation datasets. In addition, the largest R^2 value was obtained. The results show that the proposed model is the most accurate and reliable model in predicting crude oil prices. Our model effectively caught the temporal characteristics and intricate associations of time series data. The dataset spans from January 1986 to May 2023, encompassing numerous significant global economic events. These events encompassed the oil crises that occurred in the late 1980s, the global financial crisis of 2008, the oil price crash that took place in the mid-2010s, and the COVID-19 pandemic of 2020. The model enhanced the accuracy of predicting future outcomes by accurately representing changes in prices and the complex connections that arise from these occurrences.

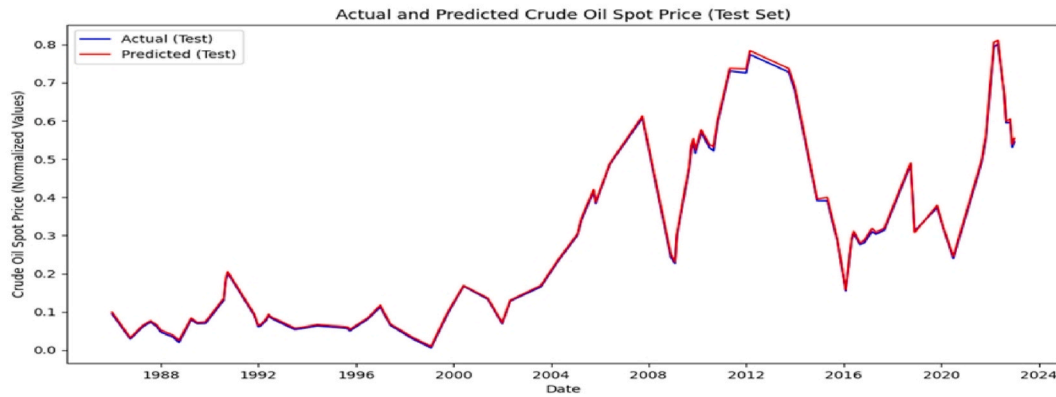
Notably, significant volatility in crude oil prices was noticed, particularly during the 2008 global financial crisis and the 2020 COVID-19 pandemic. Our model successfully captured and represented these variations and the economic mechanisms that drive them with precision. This demonstrates the model's capacity to accurately represent intricate and non-linear connections. The model correctly identified associations such as the robust positive correlation between gold prices and crude oil prices, as well as the notable negative correlation between the US 10-

Table 3
Performance metrics of models.

Model	Set	R^2	MAE	MAPE	MSE	RMSE
DDPG	Training	0.9799	0.0237	0.0685	0.0012	0.0345
	Test	0.866	0.0545	0.1468	0.0071	0.084
	15-Fold Cross Validation	0.9187	5.6837	0.1435	65.9178	7.9582
LSTM	Training	0.9236	0.0501	0.1548	0.0045	0.0673
	Test	0.8546	0.0652	0.1983	0.0077	0.0875
	15-Fold Cross Validation	0.9523	4.5785	0.1151	40.265	6.0572
SVR	Training	0.9214	0.0565	0.2102	0.0047	0.0682
	Test	0.8999	0.0559	0.2038	0.0053	0.0726
	15-Fold Cross Validation	0.9017	7.176	0.2174	76.9538	8.6952
CNN	Training	0.9177	0.0532	0.1717	0.0049	0.0698
	Test	0.8498	0.0666	0.2091	0.0079	0.0889
	15-Fold Cross Validation	0.9312	5.8006	0.1601	55.955	7.3639
XGBoost	Training	0.9679	0.0333	0.113	0.0019	0.0436
	Test	0.879	0.054	0.1567	0.0064	0.0798
	15-Fold Cross Validation	0.9346	5.2373	0.1229	55.5123	7.3222
Our Model	Training	0.9997	0.002	0.0056	0.00001	0.0042
	Test	0.9991	0.0037	0.0099	0.00004	0.007
	15-Fold Cross Validation	0.9987	0.5516	0.0119	1.2398	0.9798

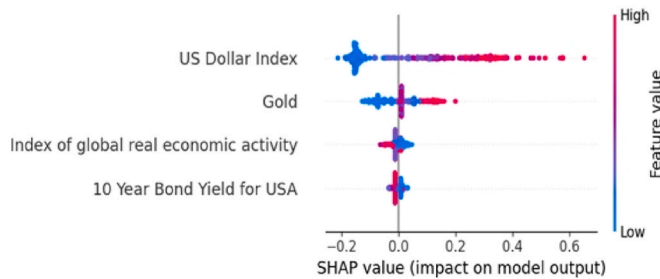


(a) Training Phase

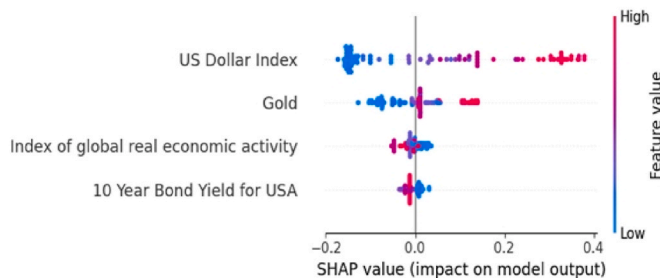


(b) Test Phase

Fig. 8. Proposed model: Training and test forecasting.



(a) Training Phase



(b) Test Phase

Fig. 9. Proposed model: Training and test data SHAP values.

year bond yield and gold. The proposed model successfully predicted the intricate economic and financial elements impacting crude oil prices and shown greater performance in comparison to other models. This showcases the model's capacity for future price forecasting and economic analysis.

Nevertheless, the utilization of SHAP analysis enabled us to get insight into the decision-making processes of the model. SHAP values quantify the individual impact of each feature on the model's prediction. The relationship between macroeconomic variables, specifically the US Dollar Index and Gold prices, and crude oil prices has been convincingly shown. SHAP research reveals that the model output is positively influenced by high values of the US Dollar Index and Gold prices, but low values have a negative impact. This demonstrates that the model has the ability to effectively capture intricate economic connections and enhances the precision of forecasts. Moreover, the utilization of the RandomSearch algorithm in this work significantly contributed to enhancing the performance of the model. RandomSearch accelerates the parameter optimization process by randomly exploring the hyperparameter space and typically yields more efficient outcomes compared to conventional approaches like GridSearch. The extensive search capacity of RandomSearch facilitated the discovery of more ideal hyperparameter combinations, hence enhancing the overall performance of our model. Hence, the utilization of the RandomSearch algorithm in our research played a crucial role in attaining exceptional precision and minimal error rates.

The study's findings demonstrate that deep learning and machine learning algorithms outperform conventional methods in forecasting crude oil prices. The exceptional performance of the suggested model specifically emphasizes its capacity to enhance forecasting accuracy by

effectively capturing the temporal characteristics and intricate associations of time series data. These results are consistent with several research in the literature. For instance, research conducted by Suriya [13] and Kulkarni & Haidar [18] has shown that artificial neural networks (ANNs) are successful in forecasting crude oil prices. Research has demonstrated that artificial neural networks (ANNs) exhibit superior accuracy in comparison to conventional econometric models. The study utilized deep learning models, including LSTM and CNN, which demonstrated satisfactory performance. However, the proposed model exhibited the highest level of accuracy. Pan et al. [17] and Hu et al. [20] have emphasized the effectiveness of artificial neural networks (ANNs) and long short-term memory (LSTM) networks in accurately representing the complex patterns of crude oil pricing. Furthermore, this study showed the efficacy of LSTM-based models in accurately capturing intricate linkages and time series properties. Specifically, our proposed model has surpassed this achievement by offering the utmost precision in its predictions. Chen et al. [27] and Zhao et al. [2] have shown that deep learning models, particularly deep reinforcement learning (DRL), outperform other methods in financial forecasting. The DDPG algorithm demonstrated superior performance in this analysis, although it did not achieve the same level of accuracy as the model proposed in this work. This demonstrates the capacity of reinforcement learning to capture non-linear correlations in intricate financial data, while also indicating that hybrid models can offer superior accuracy. Research conducted by Gao & Lei [28] and Amal [43] has demonstrated the exceptional ability of LSTM networks to effectively capture complex and nonlinear interactions within time series data. Nevertheless, our suggested model attained the utmost precision by amalgamating the temporal characteristics acquired by LSTM and the robust overall efficacy of XGBoost. In a study conducted by [41], it was demonstrated that integrating text and financial features improved the accuracy of predicting crude oil prices. This was achieved by employing a CNN-based model that utilized online media text mining. Furthermore, the combination of text and financial features outperformed the use of financial features alone in a random forest model. The study conducted by Lin et al. [51] evaluated the performance of the BiLSTM, Attention mechanism, and CNN-based WT-BLAC model combined with wavelet transform. The WT-BLAC model demonstrated the following results in WTI price prediction: RMSE of 2.2518, MAE of 1.1780, MAPE of 2.6261, and R^2 of 0.9663. It was found to have the highest accuracy in Brent price prediction, with RMSE of 1.2936, MAE of 0.8894, MAPE of 1.8027, and R^2 of 0.9900. Furthermore, Mohsin & Jamaani (2023) introduced a novel convolutional neural network (CNN) deep learning architecture. The model's results are shown as MAE with a value of 0.3245, MAPE with a value of 2.4522, and RMSE with a value of 0.4642. In a study conducted by Kim et al. (2024), the authors examined the effectiveness of RF, LightGBM, and XGBoost models for predicting oil prices. They found that the LightGBM model exhibited the highest level of prediction accuracy. The model we offered demonstrated the lowest MAE of 0.0037, MSE of 0.00004, RMSE of 0.007, and MAPE of 0.0099 across all datasets. Additionally, it yielded the highest R^2 value. Consequently, the model presented in this research is built upon the works of [41], Lin et al. [51], Mohsin & Jamaani (2023), and Kim et al. (2024), which have shown better predictive capabilities for crude oil prices. These studies have achieved much reduced error rates and greater accuracy values compared to their respective investigations. The results of this study align with prior research in the field and endorse the capacity of deep learning and hybrid models in predicting financial outcomes. The exceptional performance of the suggested model can be regarded as a potential approach for future studies in financial forecasting. Our model attained the minimum MAE, MSE, RMSE, and MAPE values across all datasets, while also yielding the highest R^2 value. Furthermore, our study's findings significantly contribute to the assessment of the model's performance during periods characterized by intense volatility, such as economic crises. For instance, the model's ability to effectively forecast price changes following from significant events like the 2008 global

financial crisis and the 2020 COVID-19 pandemic is demonstrated by the observed fluctuations. This demonstrates the model's ability to accurately capture price swings resulting from intricate economic events.

The scope of our investigation is restricted to the dataset including the timeframe from January 1986 to May 2023. The data during this time period represents certain economic circumstances and occurrences, and the model's performance may differ in other time periods and circumstances. Subsequent investigations can enhance the applicability of the concept by utilizing more extensive datasets encompassing diverse timeframes. The study incorporates the US Dollar Index, Gold Price, US 10-Year Bond Yield, and Global Economic Activity as independent variables. Nevertheless, there exist significant macroeconomic and geopolitical variables that might influence the fluctuations in crude oil prices. In order to conduct a more comprehensive evaluation of the model's effectiveness, future research should consider incorporating these additional factors.

Despite the excellent accuracy of our model, it possesses an intricate architecture and requires significant processing time. This could provide challenges in utilizing the model for real-time applications. Subsequent research could focus on creating models that use lower processing resources while maintaining comparable levels of precision. Subsequent investigations have the potential to enhance the applicability and reliability of the models by employing diverse timeframes and more extensive datasets. It is crucial to analyze the impact of certain occurrences, such as economic crises, natural catastrophes, or significant geopolitical events, in order to comprehend the performance of models under such circumstances. Moreover, incorporating other significant macroeconomic and geopolitical factors that impact crude oil prices can enhance the accuracy of the model's predictions. For instance, incorporating variables like as OPEC decisions, production limitations, and trade conflicts into the model can enhance the accuracy of the projections.

Incorporating the model with real-time data streams can offer substantial benefits in actual applications. Subsequent research could prioritize the creation of models that are capable of making real-time predictions by minimizing computing expenses. Moreover, the utilization of techniques like SHAP analysis facilitates comprehension of the decision-making processes of models and enhances the transparency of the models. Subsequent research endeavors should employ these strategies more extensively to enhance the comprehensibility of models and facilitate users' better understanding of them. Ultimately, the cooperation of specialists from many fields such as finance, economics, computer science, and statistics can enhance the creation of more thorough and efficient models. Interdisciplinary collaboration can enhance the process of developing models in future investigations. To conclude, the results produced in this study demonstrate the efficacy of deep learning and hybrid models in forecasting crude oil prices. Nevertheless, it is crucial for future research to adhere to the aforementioned suggestions in order to enhance the efficacy of these models and guarantee their applicability to a wide range of scenarios.

This study demonstrates the efficiency of deep learning and hybrid models in predicting crude oil prices. It provides a variety of recommendations for policymakers and industry experts. Encouraging the implementation of modern data analytics and artificial intelligence instruments is necessary to improve understanding and oversight of energy markets. This will enable decision makers to gain a deeper understanding of market dynamics and predict potential volatility in prices. Furthermore, improving the quality and accessibility of data is of extreme significance. Improving the accuracy and dependability of models can potentially be achieved by public institutions and the private sector through the provision of more comprehensive and current data related to crude oil markets. Additionally, including macroeconomic and geopolitical factors while designing energy policies would enhance the accuracy of price forecasts. Policymakers should augment expenditures to enhance computing power and data analytics abilities in the energy industry and fortify human resources in this field. Enhancing the

expertise and skills of energy industry personnel in data science and artificial intelligence via training programs and specialized training would improve the sector's capacity for analysis.

Moreover, the promotion of transparency and the exchange of data in energy markets is crucial to improving the development and reliability of forecasting models. Collaborations between the public and private sectors can enhance knowledge and innovation in the sector by promoting the exchange of data and the implementation of collaborative research efforts. Eventually, authorities should actively implement policies based on the results of forecasting techniques to guarantee energy security and maintain market stability. By accurately predicting the implications of OPEC decisions, trade conflicts, and other geopolitical issues, and taking effective actions in response, we can guarantee the stability and long-term viability of energy markets. For energy policy-making in this situation, it is important to use data-driven and analytical methods. Additionally, it is crucial to create policies that are flexible and adaptable to deal with future uncertainties. Implementing these suggestions will increase the effectiveness of energy market management and improve the reliability of crude oil price predictions.

Furthermore, policymakers place great importance on accurate prediction of oil prices due to its significant impact on macroeconomic indicators. Increasing oil prices are expected to have a detrimental influence on macroeconomic indicators. In this scenario, it may be possible to implement policy measures in order to maintain a balance between the total amount of goods and services available and the total amount of goods and services demanded, as a preventive step to prevent an increase in prices, particularly in countries that heavily depend on energy resources. In the end, two proposals are suggested for further research. The primary finding of this study is that the model developed, which exhibits significant advantages above the current literature, may be implemented on a country-specific level. Another aspect is that the determinant factors are primarily affected by the US dollar. These variables may be retested in the model by constructing a distinct research region, which may even be exclusive to a certain country.

CRedit authorship contribution statement

Ahmed Ihsan Simsek: Software, Methodology, Data curation. **Emre Bulut:** Investigation, Conceptualization. **Yunus Emre Gur:** Software, Data curation. **Esma Gültekin Tarla:** Investigation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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